

Assignment: Transfer Learning for Binary Forest Segmentation

Objective

The primary goal of this assignment is to conduct a comparative study assessing the efficacy and efficiency of **training from scratch** versus utilizing **transfer learning** for a specialized image segmentation task. You will be segmenting forest areas (foreground) from non-forest areas (background) using the augmented forest segmentation dataset from Kaggle.

Dataset

- **Source:** Kaggle - [Augmented Forest Segmentation](#)
- **Task:** Binary Image Segmentation (Forest vs. Non-Forest)
- **Input:** RGB Image
- **Output:** Binary Mask

Architecture and Required Experiments

You must implement a standard U-Net architecture using **ResNet50** as the pre-trained encoder. The implementation should leverage the `segmentation_models` library for ease of use.

Experiment 1: Training from Scratch (100 Epochs)

- **Encoder Weights:** Randomly initialized (`None`).
- **Training Schedule:** Train the entire model for **100 epochs**.

Experiment 2: Transfer Learning (100 Epochs - Phased Training)

- **Encoder Weights:** Initialized with **ImageNet** pre-trained weights.
- **Phase 1 (Frozen):** Train the decoder only (encoder frozen/non-trainable) for the first **50 epochs**.
- **Phase 2 (Fine-Tuning):** Unfreeze the entire model and continue training for the final **50 epochs** (total 100 epochs).

Experimental Setup (Guidelines)

You have freedom to choose hyperparameters and preprocessing techniques, but must document your choices.

- **Loss Function:** A combination loss function (e.g., Binary Cross-Entropy + Dice Loss) is highly recommended.

- **Optimizer:** Standard optimizers (e.g., Adam or SGD) are acceptable.
- **Preprocessing:** Include normalization (pixel values to $[0, 1]$) and basic resizing (e.g., 128×128) to accommodate model input requirements.

Deliverables and Analysis

Submit your assignment as a notebook containing all experiments with the outputs. Your final submission must clearly present a comparison between the two experimental models.

1. **Metric Comparison Table:** A clear table showing the final performance results for *both* models on the test set, including:
 - **Primary Metric:** Intersection over Union (IoU)
 - **Secondary Metrics:** Accuracy, Precision, and Recall
2. **Comparative Analysis:** A concise section discussing the findings, specifically:
 - **Performance:** Which model achieved higher scores across the key metrics (IoU, F1-score) and why.
 - **Efficiency:** Compare the total **training time** for the two 100-epoch runs and discuss the trade-off between performance gain and computational effort.
3. **Visual Results:** Show a side-by-side visualization for at least one sample image demonstrating:
 - Original Image
 - Ground Truth Mask
 - Prediction from Experiment 1 (Scratch Model)
 - Prediction from Experiment 2 (Transfer Learning Model)